LAPORAN PRAKTIKUM

TEKNOLOGI IOT II

Komunikasi Asinkron dan WebSockets, Topologi Jaringan Terdesentralisasi (ESP Now dan Mesh)

Disusun Oleh:

Faiz Hisyam Rasyid (25782068)

Dosen Pengajar:

Dr. Ir. Septafiansyah Dwi Putra, S.T., M.T.

Agus Ambarwari, S.Pd., M.Kom.

TEKNOLOGI REKAYASA INTERNET

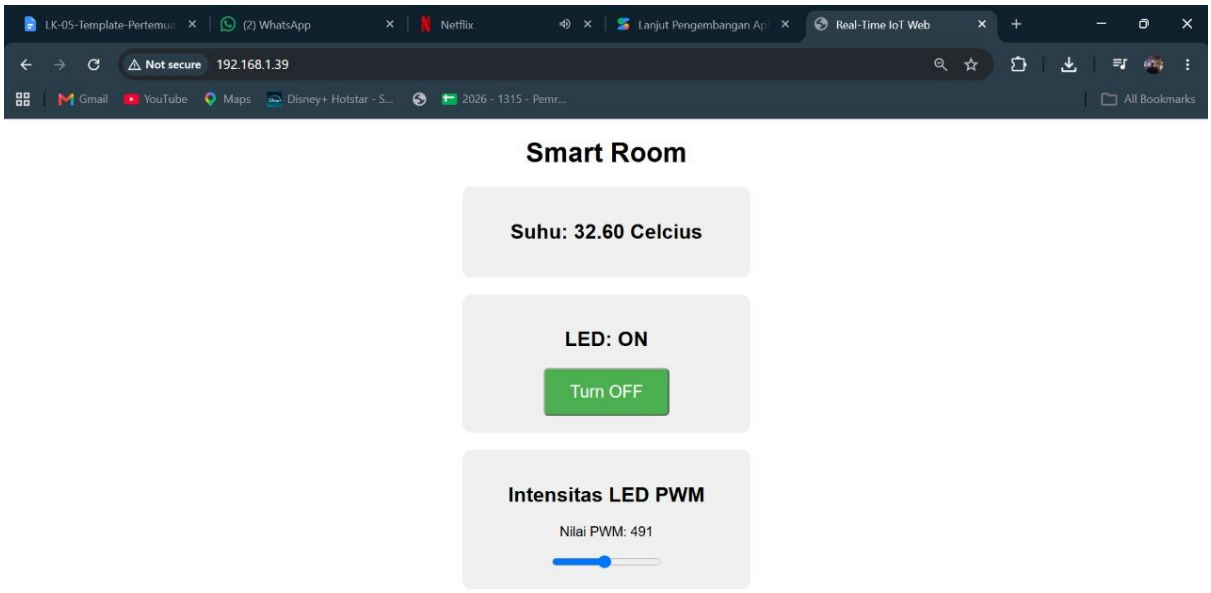
JURUSAN TEKNOLOGI INFORMASI

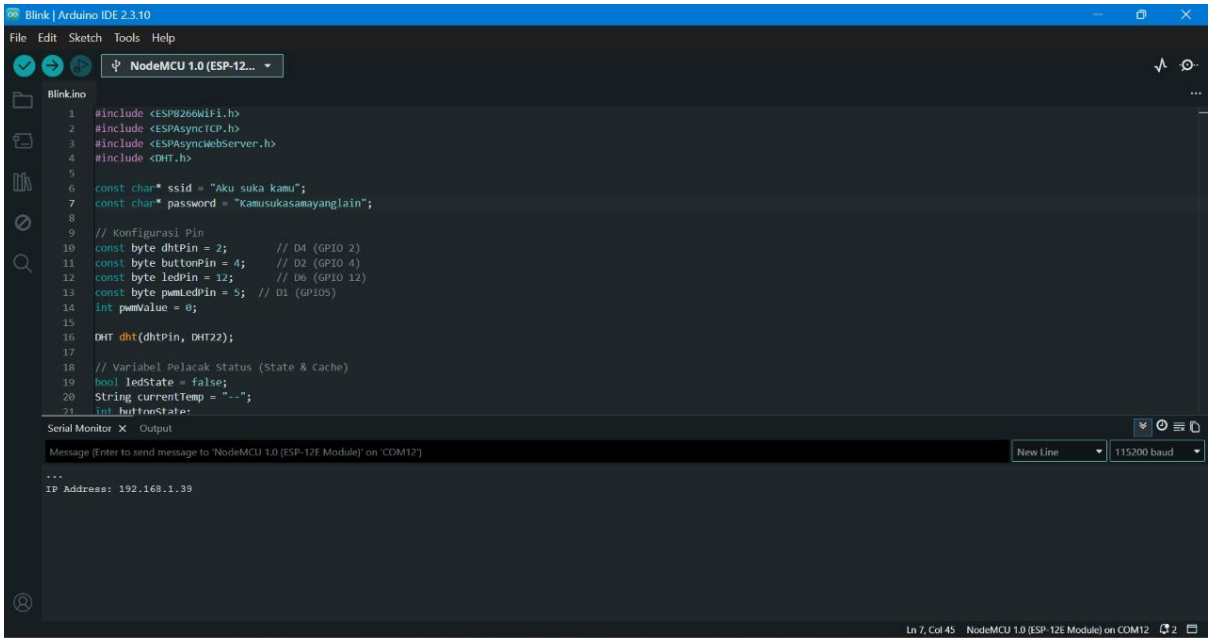
POLITEKNIK NEGERI LAMPUNG

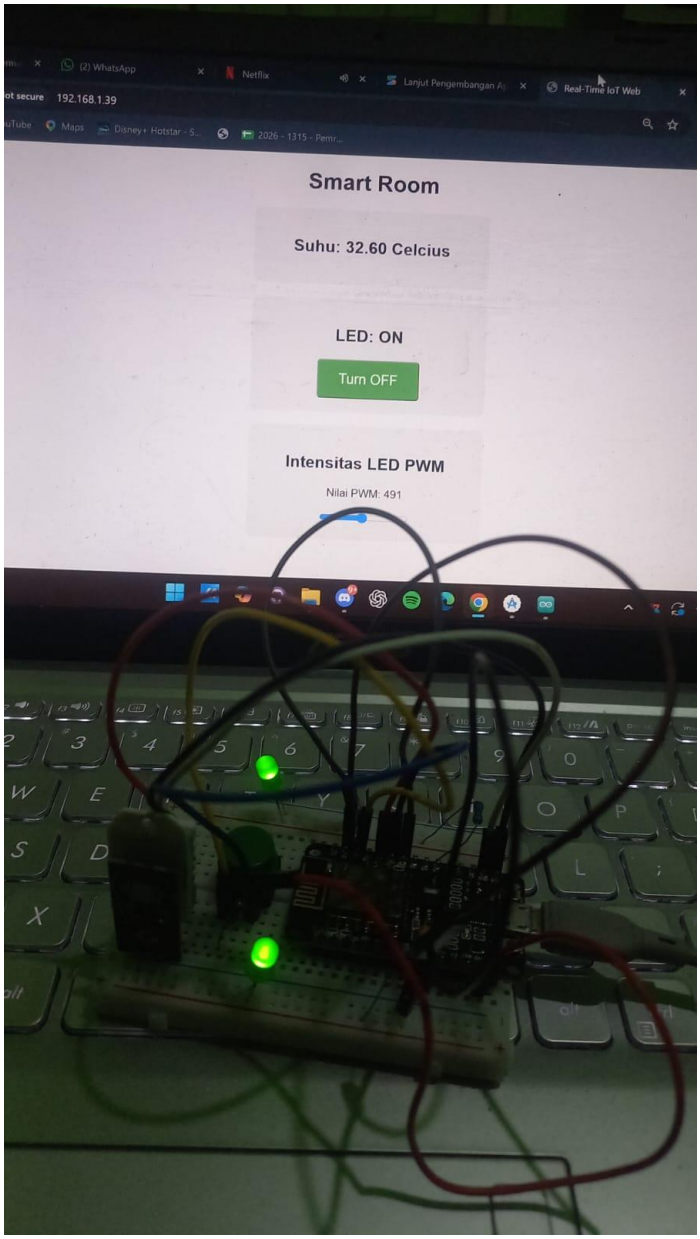
2026/2027

Modul 4: Komunikasi Asinkron dan WebSockets

Tugas



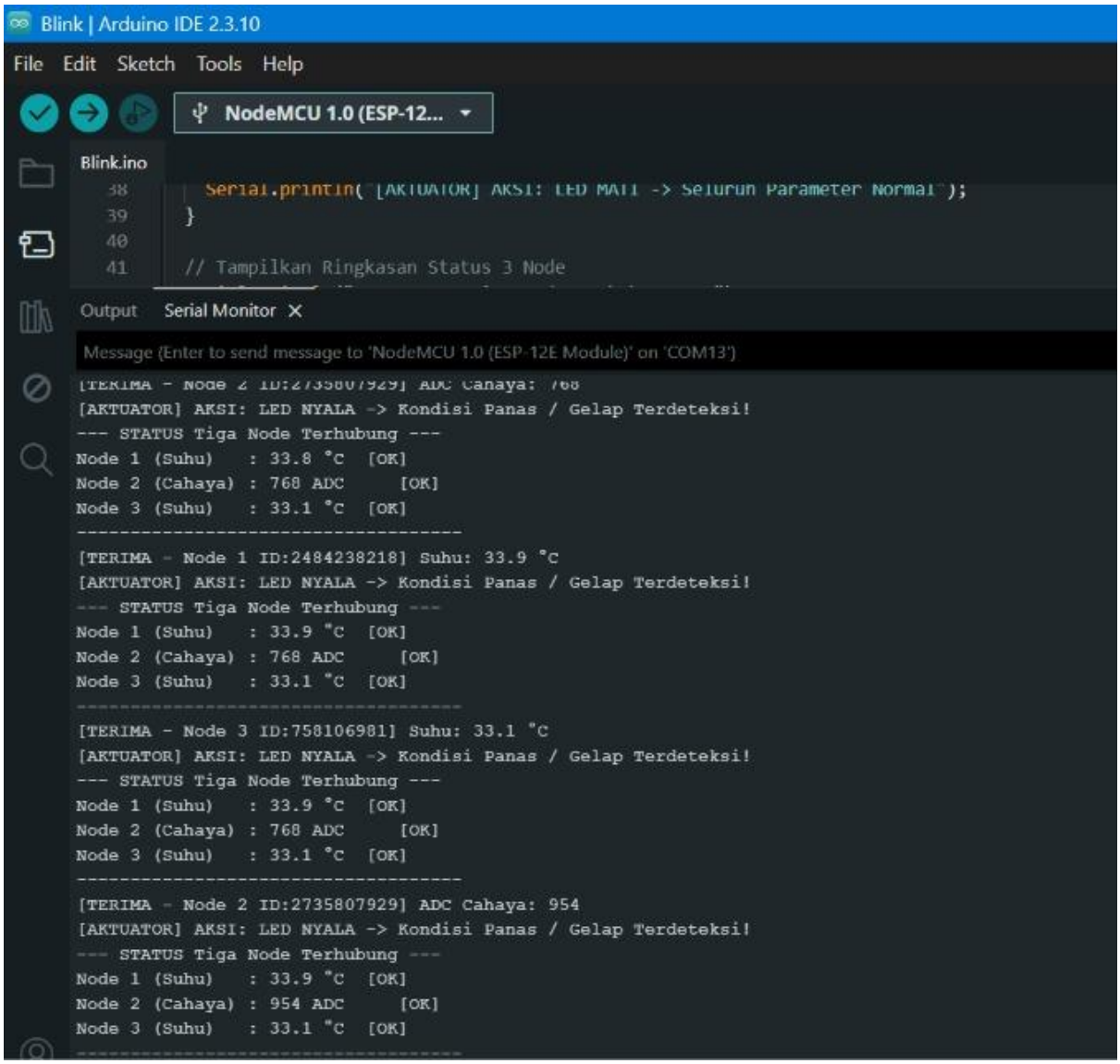



(2) WhatsApp
Netflix
Lanjut Pengembangan A
Real-Time IoT Web
ot secure 192.168.1.39
Maps
Disney+ Hotstar - S...
2026 - 1315 - Pemr...
Smart Room
Suhu: 32.60 Celcius
LED: ON
Turn OFF
Intensitas LED PWM
Nilai PWM: 491

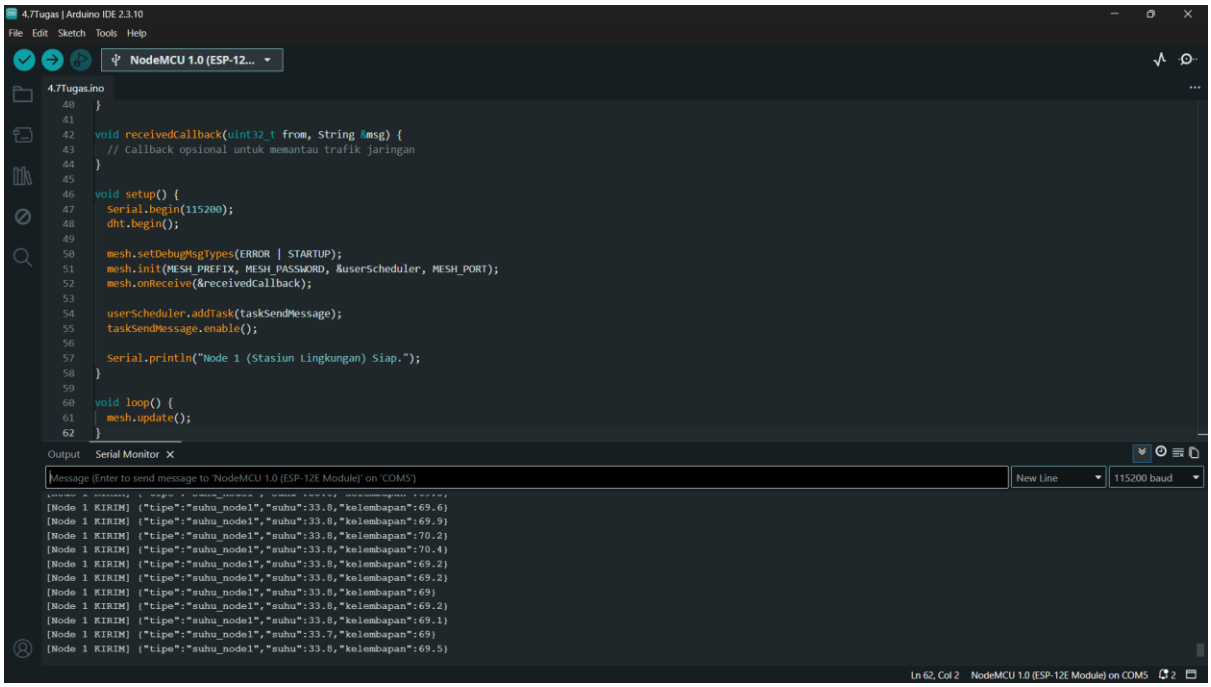
Modul 5: Topologi Jaringan Terdesentralisasi (ESP-NOW dan Mesh)

Tugas

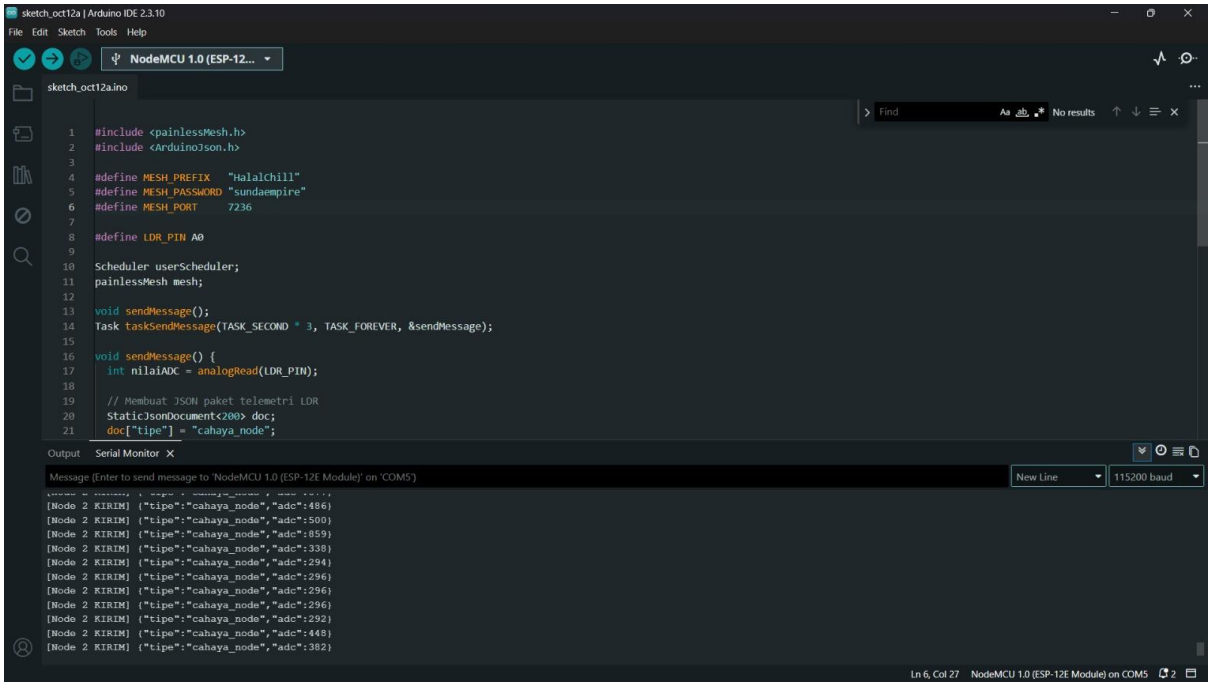
Akuator



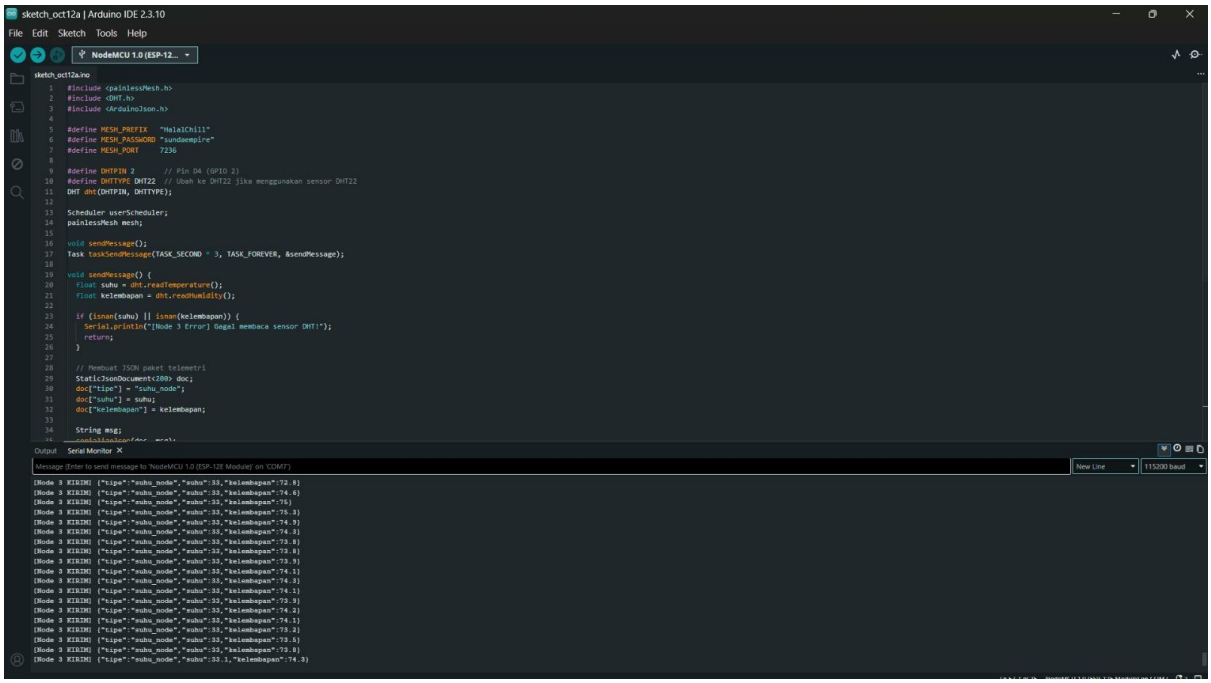
Node 1


4.7Tugas | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
4.7Tugas.ino
}
void receivedCallback(uint32_t from, String &msg) {
// Callback opsional untuk memantau trafik jaringan
}
void setup() {
Serial.begin(115200);
dht.begin();
mesh.setDebugMsgTypes(ERROR | STARTUP);
mesh.init(MESH_PREFIX, MESH_PASSWORD, &userScheduler, MESH_PORT);
mesh.onReceive(&receivedCallback);
userScheduler.addTask(taskSendMessage);
taskSendMessage.enable();
Serial.println("Node 1 (Stasiun Lingkungan) Siap.");
}
void loop() {
mesh.update();
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.6}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.9}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":70.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":70.4}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.1}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.7,"kelembapan":69}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.5}
Ln 62, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5

Node 2


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
Find
No results
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "HalalChill"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7236
#define LDR_PIN A0
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
int nilaiADC = analogRead(LDR_PIN);
// Membuat JSON paket telemetri LDR
StaticJsonDocument<200> doc;
doc["tipe"] = "cahaya_node";
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":486}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":500}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":859}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":338}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":294}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":292}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":448}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":382}
Ln 6, Col 27 NodeMCU 1.0 (ESP-12E Module) on COM5

Node 3


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <DHT.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "HalalChili"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7230
#define DHTPIN 2 // Pin D4 (GPIO 2)
#define DHTTYPE DHT22 // Ubah ke DHT22 jika menggunakan sensor DHT22
DHT dht(DHTPIN, DHTTYPE);
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
float suhu = dht.readTemperature();
float kelembapan = dht.readHumidity();
if (isnan(suhu) || isnan(kelembapan)) {
Serial.println("[Node 3 Error] Gagal membaca sensor DHT!");
return;
}
// Membuat JSON paket telemetri
StaticJsonDocument<200> doc;
doc["tipe"] = "suhu_node";
doc["suhu"] = suhu;
doc["kelembapan"] = kelembapan;
String msg;
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
New Line
115200 baud
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":72.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.6}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":76}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":75.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33.1,"kelembapan":74.3}